

A QUICK REVIEW OF TOPOLOGY

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1. FUNDAMENTAL GROUP

1.1. Homotopy. In this section we assume $I = [0, 1]$, a path in a topological space X is a continuous map $\gamma: I \rightarrow X$ and a loop is a path γ such that $\gamma(0) = \gamma(1)$.

Definition 1.1.1 (homotopy). Let X and Y be topological spaces and $f, g: X \rightarrow Y$ be continuous maps. A homotopy from f to g is a continuous map $F: X \times I \rightarrow Y$ such that for all $x \in X$, one has

$$F(x, 0) = f(x)$$

$$F(x, 1) = g(x)$$

If there exists a homotopy from f to g , then we say f and g are homotopic, and write $f \simeq g$.

Definition 1.1.2 (stationary homotopy). Let X and Y be topological spaces and $A \subseteq X$ an arbitrary subset. A homotopy F between continuous maps $f, g: X \rightarrow Y$ is said to be stationary on A if

$$F(x, t) = f(x)$$

for all $x \in A$ and $t \in I$. If there exists such a homotopy, then we say f and g are homotopic relative to A .

Remark 1.1.1. If f and g are homotopic relative to A , then f must agree with g on A .

Definition 1.1.3 (path homotopy). Let X be a topological space and γ_1, γ_2 be two paths in X . They are said to be path homotopic if they are homotopic relative on $\{0, 1\}$, and write $\gamma_1 \simeq \gamma_2$.

Definition 1.1.4 (loop homotopy). Let X be a topological space and γ_1, γ_2 be two loops in X . They're called loop homotopic if they are homotopic relative on $\{0\}$, and write $\gamma_1 \simeq \gamma_2$.

Remark 1.1.2. For convenience, if γ_1, γ_2 are paths (or loops), then when we say γ_1 is homotopic to γ_2 , we mean γ_1 is path (or loop) homotopic to γ_2 .

Definition 1.1.5 (free homotopy). Let X be a topological space and γ_1, γ_2 be two loops in X . They are said to be free (loop) homotopic if they're homotopic through loops (but not necessarily preserving the base point), that is, there exists a homotopy $F(s, t): [0, 1] \times [0, 1] \rightarrow X$ such that

$$F(s, 0) = \gamma_1(s)$$

$$F(s, 1) = \gamma_2(s)$$

$$F(0, t) = F(1, t) \text{ holds for all } t \in [0, 1]$$

1.2. Fundamental group.

Proposition 1.2.1. Let X be a topological space. For any $p, q \in X$, path homotopy is an equivalence relation on the set of all paths in X from p to q . For any path γ in X , the path homotopy class is denoted by $[\gamma]$.

Proof. For path $\gamma: I \rightarrow X$, γ is homotopic to itself by $F(s, t) = \gamma(s)$. If γ_1 is homotopic to γ_2 by F , then γ_2 is homotopic to γ_1 by $G(s, t) = F(s, 1-t)$. Finally,

suppose γ_1 is homotopic to γ_2 by F , γ_2 is homotopic to γ_3 by G . Then consider

$$H = \begin{cases} F(s, 2t) & 0 \leq t \leq \frac{1}{2} \\ G(s, 2t - 1) & \frac{1}{2} \leq t \leq 1 \end{cases}$$

which is a homotopy from γ_1 to γ_3 . This shows path homotopy is an equivalence relation. $\square$

Definition 1.2.1 (reparametrization). A reparametrization of a path $f: I \rightarrow X$ is of the form $f \circ \varphi$ for some continuous map $\varphi: I \rightarrow I$ fixing 0 and 1.

Lemma 1.2.1. Any reparametrization of a path f is homotopic to f .

Proof. Suppose $f \circ \varphi$ is a reparametrization of f , and let $F: I \times I \rightarrow I$ denote the straight-line homotopy from the identity map to φ , that is, $F(s, t) = t\varphi(s) + (1-t)s$. Then $f \circ F$ is a path homotopy from f to $f \circ \varphi$. $\square$

Definition 1.2.2 (product of path). Let X be a topological space and f, g be paths. f and g are composable if $f(1) = g(0)$. If f and g are composable, their product $f \cdot g: I \rightarrow X$ is defined by

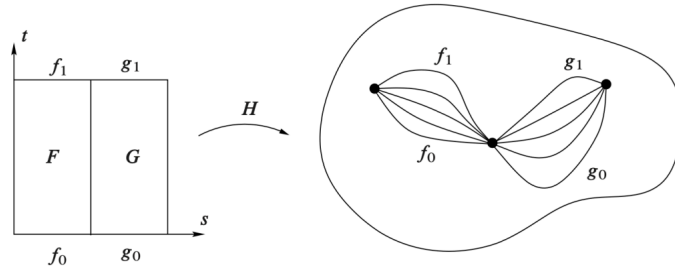
$$f \cdot g(s) = \begin{cases} f(2s) & 0 \leq s \leq \frac{1}{2} \\ g(2s - 1) & \frac{1}{2} \leq s \leq 1 \end{cases}$$

Proposition 1.2.2. Let X be a topological space and f_0, f_1, g_0, g_1 be paths in X such that f_0, g_0 are composable and f_1, g_1 are composable. If $f_0 \simeq f_1$ and $g_0 \simeq g_1$ through endpoint-preserving homotopies, then $f_0 \cdot g_0 \simeq f_1 \cdot g_1$.

Proof. Suppose the homotopy from f_0 to f_1 is given by F and the homotopy from g_0 to g_1 is given by G . Then the required homotopy H from $f_0 \cdot g_0$ to $f_1 \cdot g_1$ is given by

$$H(s, t) = \begin{cases} F(2s, t) & 0 \leq s \leq \frac{1}{2}, 0 \leq t \leq 1 \\ G(2s - 1, t) & \frac{1}{2} \leq s \leq 1, 0 \leq t \leq 1 \end{cases}$$

$\square$



Proposition 1.2.3. Let X be a topological space and f, g be paths in X such that $f \simeq g$. If $\bar{f}$ is the path obtained by reversing f , that is $\bar{f}(s) := f(1-s)$, then $\bar{f} \simeq \bar{g}$.

Proof. Suppose f is homotopic to g by homotopy F . Then $G(s, t) := F(1 - s, t)$ is a homotopy from $\bar{f}$ to $\bar{g}$ since

$$\begin{aligned} G(s, 0) &= F(1 - s, 0) = f(1 - s) = \bar{f}(s) \\ G(s, 1) &= F(1 - s, 1) = g(1 - s) = \bar{g}(s) \end{aligned}$$

□

Remark 1.2.1. With above propositions, it makes sense to define the composition of path homotopy classes by setting $[\gamma_1] \cdot [\gamma_2] := [\gamma_1 \cdot \gamma_2]$, and use the notation $[\bar{\gamma}]$.

Proposition 1.2.4. Let X be a topological space and $[f], [g], [h]$ be homotopy classes of loops based at $p \in X$.

- (1) $[c_p] \cdot [f] = [f] \cdot [c_p] = [f]$, where c_p is constant loop based at p .
- (2) $[f] \cdot [\bar{f}] = [c_p]$ and $[\bar{f}] \cdot [f] = [c_p]$.
- (3) $[f] \cdot ([g] \cdot [h]) = ([f] \cdot [g]) \cdot [h]$.

Proof. For (1). Let us show that $c_p \cdot f \simeq f$, and the other case is similar. Define $H: I \times I \rightarrow X$ by

$$H(s, t) = \begin{cases} p & t \geq 2s \\ f(\frac{2s-t}{2-t}) & t \leq 2s \end{cases}$$

This map is continuous since $f(0) = p$, and it is clear to see $H(s, 0) = f(s)$ and $H(s, 1) = c_p \cdot f(s)$. Thus H gives the desired homotopy.

For (2). It suffices to show that $f \cdot \bar{f} \simeq c_p$, since the reverse path of $\bar{f}$ is f , the other relation follows by interchanging the roles of f and $\bar{f}$. Define

$$H(s, t) = \begin{cases} f(2s) & 0 \leq s \leq \frac{t}{2} \\ f(t) & \frac{t}{2} \leq s \leq 1 - \frac{t}{2} \\ f(2-2s) & 1 - \frac{t}{2} \leq s \leq 1 \end{cases}$$

It is easy to check that H is a homotopy from c_p to $f \cdot \bar{f}$.

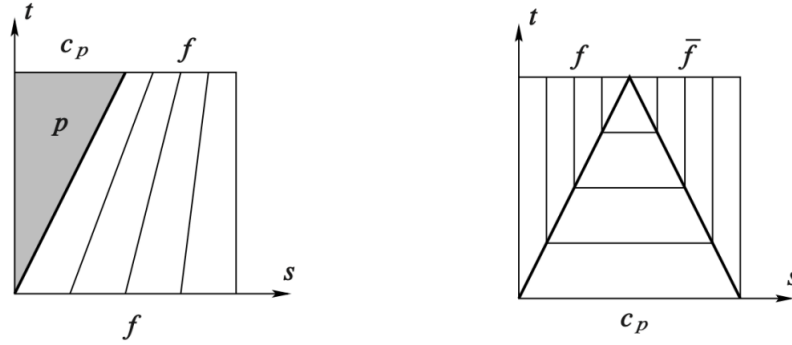
For (3). It suffices to show $(f \cdot g) \cdot h \simeq f \cdot (g \cdot h)$. The first path follows f and then g at quadruple speed for $s \in [0, \frac{1}{2}]$, and then follows h at double speed for $s \in [\frac{1}{2}, 1]$, while the second follows f at double speed and then g and h at quadruple speed. The two paths are therefore reparametrizations of each other and thus homotopic by Lemma 1.2.1.

□

Definition 1.2.3 (fundamental group). Let X be a topological space. The fundamental group of X based at $p \in X$, denoted by $\pi_1(X, p)$, is the set of path homotopy classes of loops based at p equipped with composition as its group structure.

Theorem 1.1 (base point change). Let X be a topological space, $p, q \in X$ and g is any path from p to q . The map

$$\begin{aligned} \Phi_g: \pi_1(X, p) &\rightarrow \pi_1(X, q) \\ [f] &\mapsto [\bar{g}] \cdot [f] \cdot [g] \end{aligned}$$



is a group isomorphism with inverse $\Phi_{\bar{g}}$.

Proof. It suffices to show Φ_g is a group homomorphism, since $\Phi_g \circ \Phi_{\bar{g}} = \Phi_{\bar{g}} \circ \Phi_g = \text{id}$. For $[\gamma_1], [\gamma_2] \in \pi_1(X, p)$, one has

$$\begin{aligned} \Phi_g[\gamma_1] \cdot \Phi_g[\gamma_2] &= [\bar{g}] \cdot [\gamma_1] \cdot [g] \cdot [\bar{g}] \cdot [\gamma_2] \cdot [g] \\ &= [\bar{g}] \cdot [\gamma_1] \cdot [c_p] \cdot [\gamma_2] \cdot [g] \\ &= [\bar{g}] \cdot [\gamma_1] \cdot [\gamma_2] \cdot [g] \\ &= \Phi_g([\gamma_1] \cdot [\gamma_2]) \end{aligned}$$

□

Corollary 1.1. If X is path-connected, then its fundamental groups at different base points are isomorphic, although the isomorphism is not canonical without choosing a path. We write $\pi_1(X)$ when the base point is understood.

Definition 1.2.4. If X is a path-connected topological space with $\pi_1(X) = 0$, then it is called simply-connected.

Theorem 1.2. The fundamental group of a topological manifold M is countable.

Proof. Since M is second countable, there exists a countable cover $\mathcal{U}$ of M consisting of coordinate balls, and for each $U, U' \in \mathcal{U}$ the intersection $U \cap U'$ has at most countably many components. We choose a point in each such component and let $\mathcal{X}$ denote the (countable) set consisting of all the chosen points as U, U' range over all the sets in $\mathcal{U}$. For each $U \in \mathcal{U}$ and $x, x' \in \mathcal{X}$ such that $x, x' \in U$, choose a definite path $h_{x, x'}^U$ from x to x' in U .

Now choose any point $p \in \mathcal{X}$ as base point. Let us say that a loop based at p is special if it is a finite product of paths of the form $h_{x, x'}^U$. Because both $\mathcal{U}$ and $\mathcal{X}$ are countable sets, there are only countably many special loops. Each special loop determines an element of $\pi_1(M, p)$. If we can show that every element of $\pi_1(M, p)$ is obtained in this way, we are done, because we will have exhibited a surjective map from a countable set onto $\pi_1(M, p)$.

So suppose f is any loop based at p . By the Lebesgue number lemma there is an integer n such that f maps each subinterval $[(k-1)/n, k/n]$ into one of the balls in $\mathcal{U}$, which is called U_k . Let $f_k = f|_{[(k-1)/n, k/n]}$ reparametrized on the unit interval, so that $[f] = [f_1] \cdots [f_n]$.

For each $k = 1, \dots, n-1$, the point $f(k/n)$ lies in $U_k \cap U_{k+1}$. Therefore, there is some $x_k \in \mathcal{X}$ that lies in the same component of $U_k \cap U_{k+1}$ as $f(k/n)$. Choose a path g_k in $U_k \cap U_{k+1}$ from x_k to $f(k/n)$, and set $\tilde{f}_k = g_{k-1} \cdot f_k \cdot \bar{g}_k$ (taking $x_k = p$ and g_k to be the constant path c_p when $k = 0$ or n). It is immediate that $[f] = [\tilde{f}_1] \cdots [\tilde{f}_n]$, because all the g_k 's cancel out. But for each k , $\tilde{f}_k$ is a path in U_k from x_{k-1} to x_k , and since U_k is simply connected, $\tilde{f}_k$ is path-homotopic to $h_{x_{k-1}x_k}^{U_k}$. This shows that f is path-homotopic to a special loop and completes the proof. $\square$

2. COVERING SPACE

In this section, we assume¹ that all topological spaces are connected and locally path-connected, and that all maps are continuous. References for this section are [Hat02] and [Lee10].

Definition 2.0.1 (covering space). A covering space of X is a map $\pi: \tilde{X} \rightarrow X$ such that there exists a discrete space D and for each $x \in X$ an open neighborhood $U \subseteq X$, such that $\pi^{-1}(U) = \coprod_{d \in D} V_d$ and $\pi|_{V_d}: V_d \rightarrow U$ is a homeomorphism for each $d \in D$.

- (1) Such a U is called evenly covered by $\{V_d\}$.
- (2) The open sets $\{V_d\}$ are called sheets.
- (3) For each $x \in X$, the discrete subset $\pi^{-1}(x)$ is called the fiber of x .
- (4) The degree of the covering is the cardinality of the space D .

Definition 2.0.2 (isomorphism between covering spaces). Let $\pi_1: \tilde{X}_1 \rightarrow X$ and $\pi_2: \tilde{X}_2 \rightarrow X$ be two covering spaces. An isomorphism between covering spaces is a homeomorphism $f: \tilde{X}_1 \rightarrow \tilde{X}_2$ such that $\pi_1 = \pi_2 \circ f$.

2.1. Proper map.

Definition 2.1.1 (proper). Let $f: X \rightarrow Y$ be a continuous map between topological spaces. f is called proper if preimage of any compact set in Y is a compact subset in X .

Lemma 2.1.1. Let $p: X \rightarrow Y$ be a proper map between topological spaces and Y be locally compact and Hausdorff. Then p is a closed map.

Proof. Let C be a closed subset of X . We need to prove that $p(C)$ is closed in Y , that is, that $Y \setminus p(C)$ is open. Let $y \in Y \setminus p(C)$. Then y has a compact neighborhood V since Y is locally compact. The set $p^{-1}(V)$ is compact since p is proper. Let $E = C \cap p^{-1}(V)$. Then E , and hence $p(E)$, is compact. The set $p(E)$ is closed because Y is Hausdorff. Thus $U = V \setminus p(E)$ is a neighborhood of y disjoint from $p(C)$, as desired. $\square$

Corollary 2.1. Let $p: X \rightarrow Y$ be a proper map between topological spaces and Y be locally compact and Hausdorff. If $y \in Y$ and V is an open neighborhood of $p^{-1}(y)$, then there exists an open neighborhood U of y with $p^{-1}(U) \subseteq V$.

Proof. Since V is open, one has $X \setminus V$ is closed, and thus $A := p(X \setminus V)$ is also closed with $y \notin A$ since p is a closed map by Lemma 2.1.1. Thus $U := Y \setminus A$ is an open neighborhood of y such that $p^{-1}(U) \subseteq V$. $\square$

Theorem 2.1. Let $p: X \rightarrow Y$ be a surjective proper local homeomorphism between topological spaces, where Y is locally compact and Hausdorff. Then p is a covering map.

Proof. For $y \in Y$, one has $\{y\}$ is a compact set since Y is locally compact and Hausdorff, and hence so is $p^{-1}(y)$ since p is proper. On the other hand, $p^{-1}(y)$ is a discrete set since p is a local homeomorphism. Then $p^{-1}(y)$ is a finite

¹We include these hypotheses because most of the results below, such as the lifting criterion, require them, and common examples such as connected manifolds satisfy them. A connected, locally path-connected space is path-connected.

set, and we denote it by $\{x_1, \dots, x_n\}$. Since p is a local homeomorphism, for each $i = 1, \dots, n$, there exists an open neighborhood W_i of x_i and an open neighborhood U_i of y such that $p|_{W_i}$ is a homeomorphism. without loss of generality we may assume W_i are pairwise disjoint. Now $W_1 \cup \dots \cup W_n$ is an open neighborhood of $p^{-1}(y)$. Thus by Corollary 2.1 there exists an open neighborhood $U \subseteq U_1 \cap \dots \cap U_n$ of y with $p^{-1}(U) \subseteq W_1 \cup \dots \cup W_n$. If we let $V_i = W_i \cap p^{-1}(U)$, then the V_i are disjoint open sets with

$$p^{-1}(U) = V_1 \cup \dots \cup V_n$$

and all the mappings $p|_{V_i}$ are homeomorphisms. This shows p is a covering map. $\square$

2.2. The lifting theorems.

Proposition 2.2.1 (unique lifting property). Let $\pi: \tilde{X} \rightarrow X$ be a covering space, let Y be connected, and let $f: Y \rightarrow X$ be a map. If two lifts $\tilde{f}_1, \tilde{f}_2: Y \rightarrow \tilde{X}$ of f agree at one point of Y , then $\tilde{f}_1$ and $\tilde{f}_2$ agree on all of Y .

Proof. Let A be the set of points of Y where $\tilde{f}_1$ and $\tilde{f}_2$ agree. It is nonempty by hypothesis. If $y \notin A$, choose an evenly covered neighborhood of $f(y)$ and let $\tilde{U}_1$ and $\tilde{U}_2$ be the distinct sheets containing $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$, respectively. By continuity, some neighborhood N of y is mapped into these two disjoint sheets, so the lifts disagree throughout N . Hence A is closed. If $y \in A$, the two lifts land at the same point of a single sheet $\tilde{U}$. After shrinking a neighborhood N of y , both lifts map N into $\tilde{U}$, where they both equal $(\pi|_{\tilde{U}})^{-1} \circ f$. Thus A is also open, and $A = Y$ because Y is connected. $\square$

Theorem 2.2 (homotopy lifting property). Let $\pi: \tilde{X} \rightarrow X$ be a covering space and $F: Y \times I \rightarrow X$ be a homotopy. If there exists a map $\tilde{F}: Y \times \{0\} \rightarrow \tilde{X}$ which lifts $F|_{Y \times \{0\}}$, then there exists a unique homotopy $\tilde{F}: Y \times I \rightarrow \tilde{X}$ which lifts F and restricting to the given $\tilde{F}$ on $Y \times \{0\}$. Furthermore, if F is stationary on A , so is $\tilde{F}$.

Proof. Firstly, let's construct a lift $\tilde{F}: N \times I \rightarrow \tilde{X}$ for some neighborhood N in Y of a given point $y_0 \in Y$. Since F is continuous, every point $(y_0, t) \in Y \times I$ has a product neighborhood $N_t \times (a_t, b_t)$ such that $F(N_t \times (a_t, b_t))$ is contained in an evenly covered neighborhood of $F(y_0, t)$. By compactness of $\{y_0\} \times I$, finitely many such products $N_t \times (a_t, b_t)$ cover $\{y_0\} \times I$. This implies that we can choose a single neighborhood N of y_0 and a partition $0 = t_0 < t_1 < \dots < t_m = 1$ of I such that for each i , one has $F(N \times [t_i, t_{i+1}])$ is contained in an evenly covered neighborhood U_i . Suppose $\tilde{F}$ has been constructed on $N \times [0, t_i]$, starting with the given $\tilde{F}$ on $N \times \{0\}$. Since U_i is evenly covered, there is an open set $\tilde{U}_i$ of $\tilde{X}$ projecting homeomorphically onto U_i by π and containing the point $\tilde{F}(y_0, t_i)$. After replacing N by a smaller neighborhood of y_0 we may assume that $\tilde{F}(N \times \{t_i\})$ is contained in $\tilde{U}_i$. Now we can define $\tilde{F}$ on $N \times [t_i, t_{i+1}]$ to be the composition of F with the homeomorphism $\pi^{-1}: U_i \rightarrow \tilde{U}_i$ since $F(N \times [t_i, t_{i+1}]) \subseteq U_i$. After a finite number of steps we eventually get a lift $\tilde{F}: N \times I \rightarrow \tilde{X}$ for some neighborhood N of y_0 .

Next we show the uniqueness part in the special case that Y is a point, since in this case we can omit Y from the notation. Suppose $\tilde{F}$ and $\tilde{F}'$ are two lifts of $F: I \rightarrow X$ such that $\tilde{F}(0) = \tilde{F}'(0)$. As before, choose a partition

$0 = t_0 < t_1 < \cdots < t_m = 1$ of I so that for each i , one has $F([t_i, t_{i+1}])$ is contained in some evenly covered neighborhood U_i . Assume inductively that $\tilde{F} = \tilde{F}'$ on $[0, t_i]$. Since $[t_i, t_{i+1}]$ is connected, so is $\tilde{F}([t_i, t_{i+1}])$, which must therefore lie in a single one of the disjoint open sets $\tilde{U}_i$ projecting homeomorphically to U_i . Similarly, $\tilde{F}'([t_i, t_{i+1}])$ lies in a single $\tilde{U}_i$, in fact in the same one that contains $\tilde{F}([t_i, t_{i+1}])$ since $\tilde{F}'(t_i) = \tilde{F}(t_i)$. Because π is injective on $\tilde{U}_i$ and $\pi \circ \tilde{F} = \pi \circ \tilde{F}'$, it follows that $\tilde{F} = \tilde{F}'$ on $[t_i, t_{i+1}]$, and the induction step is finished.

The last step in the proof of is to observe that since the $\tilde{F}$ constructed above on sets of the form $N \times I$ are unique when restricted to each segment $\{y\} \times I$, they must agree whenever two such sets $N \times I$ overlap. So we obtain a well-defined lift $\tilde{F}$ on all of $Y \times I$. This $\tilde{F}$ is continuous since it is continuous on each $N \times I$, and $\tilde{F}$ is unique since it is unique on each segment $\{y\} \times I$. $\square$

Corollary 2.2 (path lifting property). Let $\pi: \tilde{X} \rightarrow X$ be a covering space. Suppose $\gamma: I \rightarrow X$ is any path, and $\tilde{x} \in \tilde{X}$ is any point in the fiber of $\pi^{-1}(\gamma(0))$. Then there exists a unique lift $\tilde{\gamma}: I \rightarrow \tilde{X}$ of γ such that $\tilde{\gamma}(0) = \tilde{x}$.

Proof. Let Y be a point and F be the path γ in Theorem 2.2. $\square$

Corollary 2.3 (monodromy theorem). Let $\pi: \tilde{X} \rightarrow X$ be a covering space. Suppose γ_1 and γ_2 are paths in X which are homotopic, and $\tilde{\gamma}_1, \tilde{\gamma}_2$ are their lifts with the same initial point. Then $\tilde{\gamma}_1$ is homotopic to $\tilde{\gamma}_2$.

Proof. Suppose $F: I \times I \rightarrow X$ is the homotopy from γ_1 to γ_2 which is stationary on $\{0, 1\}$ and $\tilde{\gamma}_1, \tilde{\gamma}_2$ are lifts of γ_1, γ_2 with the same initial point. Then by Theorem 2.2 there exists a homotopy $\tilde{F}: I \times I \rightarrow \tilde{X}$ from $\tilde{\gamma}_1$ to $\tilde{\gamma}_2$ which is also stationary on $\{0, 1\}$, which shows $\tilde{\gamma}_1$ is homotopic to $\tilde{\gamma}_2$. $\square$

Corollary 2.4. Let $\pi: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space. Then

- (1) The map $\pi_*: \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$ is injective.
- (2) $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ consists of the homotopy class of loops in X whose lifts to $\tilde{X}$ are still loops.
- (3) The index of $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ in $\pi_1(X, x_0)$ is the degree of covering. In particular, the degree of universal covering equals $|\pi_1(X, x_0)|$.

Proof. For (1). An element of $\ker \pi_*$ is represented by a loop $\tilde{\gamma}_0: I \rightarrow \tilde{X}$ with a homotopy F of $\gamma_0 = \pi \circ \tilde{\gamma}_0$ to the trivial loop γ_1 . By Theorem 2.2 there is a lifted homotopy of loops $\tilde{F}$ starting with $\tilde{\gamma}_0$ and ending with a constant loop. Hence $[\tilde{\gamma}_0] = 0$ in $\pi_1(\tilde{X}, \tilde{x}_0)$ and π_* is injective.

For (2). The loops at x_0 lifting to loops at $\tilde{x}_0$ certainly represent elements of the image of $\pi_*: \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$. Conversely, a loop representing an element of the image of π_* is homotopic to a loop having such a lift, so by Theorem 2.2, the loop itself must have such a lift.

For (3). For a loop γ in X based at x_0 , let $\tilde{\gamma}$ be its lift to $\tilde{X}$ starting at $\tilde{x}_0$. A product $h \cdot \gamma$ with $[h] \in H = \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ has the lift $\tilde{h} \cdot \tilde{\gamma}$ ending at the same point as $\tilde{\gamma}$ since $\tilde{h}$ is a loop. Thus we may define a function Φ from cosets $H[\gamma]$ to $\pi^{-1}(x_0)$ by sending $H[\gamma]$ to $\tilde{\gamma}(1)$. The path-connectedness of $\tilde{X}$ implies that Φ is surjective since $\tilde{x}_0$ can be joined to any point in $\pi^{-1}(x_0)$ by a path $\tilde{\gamma}$ projecting to a loop γ at x_0 . To see that Φ is injective, observe that $\Phi(H[\gamma_1]) = \Phi(H[\gamma_2])$ implies that $\gamma_1 \cdot \bar{\gamma}_2$ lifts to a loop in $\tilde{X}$ based at $\tilde{x}_0$, so $[\gamma_1][\gamma_2]^{-1} \in H$ and hence $H[\gamma_1] = H[\gamma_2]$. Thus the index of H is the same as $|\pi^{-1}(x_0)|$, which is the degree of the covering. $\square$

Proposition 2.2.2 (lifting criterion). Let $\pi: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space and $f: (Y, y_0) \rightarrow (X, x_0)$ be a map. A lift $\tilde{f}: (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$ of f exists if and only if $f_*(\pi_1(Y, y_0)) \subseteq \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$.

Proof. The only if" statement is obvious since $f_* = \pi_* \circ f_*$. Conversely, let $y \in Y$ and let γ be a path in Y from y_0 to y . By Corollary 2.2, the path $f\gamma$ in X starting at x_0 has a unique lift $\tilde{f}\gamma$ starting at $\tilde{x}_0$, and we define $\tilde{f}(y) = \tilde{f}\gamma(1)$.

To see it is well-defined, let γ' be another path from y_0 to y . Then $(f\gamma') \cdot (\overline{f\gamma})$ is a loop h_0 at x_0 with $[h_0] \in f_*(\pi_1(Y, y_0)) \subseteq \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$. This means there is a homotopy H of h_0 to a loop h_1 that lifts to a loop $\tilde{h}_1$ in $\tilde{X}$ based at $\tilde{x}_0$. Apply Theorem 2.2 to H to get a lifting $\tilde{H}$. Since $\tilde{h}_1$ is a loop at $\tilde{x}_0$, so is $\tilde{h}_0$. By Proposition 2.2.1, that is uniqueness of lifted paths, the first half of $\tilde{h}_0$ is $\tilde{f}\gamma'$ and the second half is $\tilde{f}\gamma$ traversed backwards, with the common midpoint $\tilde{f}\gamma(1) = \tilde{f}\gamma'(1)$. This shows $\tilde{f}$ is well-defined.

To see $\tilde{f}$ is continuous, let $U \subseteq X$ be an evenly covered open neighborhood of $f(y)$, and let $\tilde{U} \subseteq \tilde{X}$ be the sheet containing $\tilde{f}(y)$. Choose a path-connected open neighborhood V of y with $f(V) \subseteq U$. For paths from y_0 to points $y' \in V$, we can take a fixed path γ from y_0 to y followed by paths η in V from y to y' . Then $(f\gamma) \cdot (f\eta)$ lifts to $(\tilde{f}\gamma) \cdot (\tilde{f}\eta)$, where $\tilde{f}\eta = (\pi|_{\tilde{U}})^{-1} \circ f\eta$. Thus $\tilde{f}(V) \subseteq \tilde{U}$ and $\tilde{f}|_V = (\pi|_{\tilde{U}})^{-1} \circ f|_V$, so $\tilde{f}$ is continuous at y . $\square$

Corollary 2.5. Let $\pi: \tilde{X} \rightarrow X$ be a covering space and Y be a simply-connected space. Then every map $f: Y \rightarrow X$ has a lift.

Proof. It is clear $f_*(\pi_1(Y, y_0)) \subseteq \pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ since $\pi_1(Y, y_0) = 0$. $\square$

Theorem 2.3. Suppose X is a topological manifold, E is a Hausdorff space and $\pi: E \rightarrow X$ is a local homeomorphism with the path lifting property. Then π is a covering space.

Proof. See Theorem 4.19 of [For91]. $\square$

2.3. The classification of the covering spaces.

Definition 2.3.1 (universal covering). A simply-connected covering space of X is called universal covering.

Definition 2.3.2 (semilocally simply-connected). A topological space X is called semilocally simply-connected if each $x \in X$ has a neighborhood U such that the inclusion induced map $\pi_1(U, x) \rightarrow \pi_1(X, x)$ is trivial.

Theorem 2.4. If X is a semilocally simply-connected topological space, then X has a universal covering $\tilde{X}$.

Proof. See construction in page 64 of [Hat02]. $\square$

Proposition 2.3.1. Suppose X is a semilocally simply-connected topological space. Then for every subgroup $H \subseteq \pi_1(X, x_0)$, there exists a covering space $\pi: X_H \rightarrow X$ such that $\pi_*(\pi_1(X_H, \tilde{x}_0)) = H$ for a suitably chosen based point $\tilde{x}_0 \in X_H$.

Proof. See Proposition 1.36 of [Hat02]. $\square$

Lemma 2.3.1. Let $\pi_1: \tilde{X}_1 \rightarrow X$ and $\pi_2: \tilde{X}_2 \rightarrow X$ be two coverings. There exists an isomorphism $f: \tilde{X}_1 \rightarrow \tilde{X}_2$ taking a basepoint $\tilde{x}_1 \in \pi_1^{-1}(x_0)$ to a basepoint $\tilde{x}_2 \in \pi_2^{-1}(x_0)$ if and only if $\pi_{1*}(\pi_1(\tilde{X}_1, \tilde{x}_1)) = \pi_{2*}(\pi_1(\tilde{X}_2, \tilde{x}_2))$.

Proof. If there is an isomorphism $f: (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$, then from the two relations $\pi_1 = \pi_2 \circ f$ and $\pi_2 = \pi_1 \circ f^{-1}$ it follows that $\pi_{1*}(\pi_1(\tilde{X}_1, \tilde{x}_1)) = \pi_{2*}(\pi_1(\tilde{X}_2, \tilde{x}_2))$. Conversely, suppose that $\pi_{1*}(\pi_1(\tilde{X}_1, \tilde{x}_1)) = \pi_{2*}(\pi_1(\tilde{X}_2, \tilde{x}_2))$. By Proposition 2.2.2, that is lifting criterion, we may lift π_1 to a map $\tilde{\pi}_1: (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$ with $\pi_2 \circ \tilde{\pi}_1 = \pi_1$. Similarly, one has $\tilde{\pi}_2: (\tilde{X}_2, \tilde{x}_2) \rightarrow (\tilde{X}_1, \tilde{x}_1)$ with $\pi_1 \circ \tilde{\pi}_2 = \pi_2$. Then Proposition 2.2.1, that is the unique lifting property, $\tilde{\pi}_1 \circ \tilde{\pi}_2 = \text{id}$ and $\tilde{\pi}_2 \circ \tilde{\pi}_1 = \text{id}$ since these composed lifts fix the basepoints. $\square$

Lemma 2.3.2. For covering $\pi: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$, changing the basepoint $\tilde{x}_0$ within $\pi^{-1}(x_0)$ corresponds exactly to changing $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ to a conjugate subgroup of $\pi_1(X, x_0)$.

Proof. Let $\tilde{x}_1$ be another basepoint in $\pi^{-1}(x_0)$ and $\tilde{\gamma}$ be a path from $\tilde{x}_0$ to $\tilde{x}_1$. Then $\tilde{\gamma}$ projects to a loop γ in X representing some element $g \in \pi_1(X, x_0)$. If we denote $H_i = \pi_*(\pi_1(\tilde{X}, \tilde{x}_i))$ for $i = 0, 1$, there is an inclusion $g^{-1}H_0g \subseteq H_1$ since if $\tilde{f}$ is a loop at $\tilde{x}_0$, one has $\tilde{\gamma}^{-1} \cdot \tilde{f} \cdot \tilde{\gamma}$ is a loop at $\tilde{x}_1$. Similarly one has $gH_1g^{-1} \subseteq H_0$. This shows changing the basepoint from $\tilde{x}_0$ to $\tilde{x}_1$ changes H_0 to the conjugate subgroup $H_1 = g^{-1}H_0g$. $\square$

Theorem 2.5. Let X be a semilocally simply-connected topological space. Then there is a bijection between the set of basepoint-preserving isomorphism classes of covering spaces $\pi: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and the set of subgroups of $\pi_1(X, x_0)$ obtained by associating the subgroup $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0))$ to the covering space $(\tilde{X}, \tilde{x}_0)$. If basepoints are ignored, this correspondence gives a bijection between isomorphism classes of covering spaces $\pi: \tilde{X} \rightarrow X$ and conjugacy classes of subgroups of $\pi_1(X, x_0)$.

Proof. Proposition 2.3.1 and Lemma 2.3.1 completes the proof of the first half, and Lemma 2.3.2 completes the proof of the last half. $\square$

Corollary 2.6. Let X be a semilocally simply-connected topological space. Then the universal covering of X is unique up to isomorphism.

2.4. The structure of the deck transformation group.

Definition 2.4.1 (deck transformation). Let $\pi: \tilde{X} \rightarrow X$ be a covering space. The deck transformation group is following set

$$\text{Aut}_\pi(\tilde{X}) = \{f: \tilde{X} \rightarrow \tilde{X} \text{ is homeomorphism} \mid \pi \circ f = \pi\}$$

equipped with composition as group operation.

Proposition 2.4.1. Let $\pi: \tilde{X} \rightarrow X$ be a covering space. The deck transformation group $\text{Aut}_\pi(\tilde{X})$ acts on $\tilde{X}$ freely.

Proof. Suppose $f: \tilde{X} \rightarrow \tilde{X}$ is a deck transformation admitting a fixed point. Since $\pi \circ f = \pi$, we may regard f as a lift of π , and identity map of $\tilde{X}$ is another lift of π . By Proposition 2.2.1, that is unique lifting property, one has f is exactly identity map since it agrees with identity map at fixed point. $\square$

Definition 2.4.2 (normal). A covering $\pi: \tilde{X} \rightarrow X$ is called normal if its deck transformation group acts transitively on every fiber $\pi^{-1}(x)$.

Proposition 2.4.2. If $\pi: \tilde{X} \rightarrow X$ is a normal covering, then $\tilde{X}/\text{Aut}_\pi(\tilde{X})$ is homeomorphic to X .

Proof. Let $\Phi: \tilde{X}/\text{Aut}_\pi(\tilde{X}) \rightarrow X$ be the map sending the orbit $\mathcal{O}_{\tilde{x}}$ to $\pi(\tilde{x})$, where $\tilde{x} \in \tilde{X}$. It is clear Φ is well-defined bijection since $\text{Aut}_\pi(\tilde{X})$ acts on $\tilde{X}$ fiberwise transitive, and the following diagram commutes

$$\begin{array}{ccc} \tilde{X} & & \\ p \downarrow & \searrow \pi & \\ \tilde{X}/\text{Aut}_\pi(\tilde{X}) & \xrightarrow{\Phi} & X \end{array}$$

This diagram shows Φ is both continuous and open, since p is the quotient map and π is continuous and open, which shows $\tilde{X}/\text{Aut}_\pi(\tilde{X})$ is homeomorphic to X . $\square$

Proposition 2.4.3. Let $\pi: (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a covering space and $H = \pi_*(\pi_1(\tilde{X}, \tilde{x}_0)) \subseteq \pi_1(X, x_0)$. Then

- (1) π is a normal covering if and only if H is a normal subgroup of $\pi_1(X, x_0)$.
- (2) $\text{Aut}_\pi(\tilde{X})$ is isomorphic to the quotient $N(H)/H$, where $N(H)$ is the normalizer of H in $\pi_1(X, x_0)$.

In particular, $\text{Aut}_\pi(\tilde{X}) \cong \pi_1(X, x_0)$ if $\tilde{X}$ is universal covering.

Proof. For (1). By proof of Lemma 2.3.2 one has changing the basepoint $\tilde{x}_0 \in \pi^{-1}(x_0)$ to $\tilde{x}_1 \in \pi^{-1}(x_0)$ corresponds precisely to conjugating H by an element $[\gamma] \in \pi_1(X, x_0)$ where γ lifts to a path $\tilde{\gamma}$ from $\tilde{x}_0$ to $\tilde{x}_1$. Thus $[\gamma]$ is in the normalizer $N(H)$ if and only if $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0)) = \pi_*(\pi_1(\tilde{X}, \tilde{x}_1))$, which is equivalent to the existence of a deck transformation taking $\tilde{x}_0$ to $\tilde{x}_1$ by Lemma 2.3.1. Thus the covering space is normal if and only if $N(H) = \pi_1(X, x_0)$, that is, $H \subseteq \pi_1(X, x_0)$ is a normal subgroup.

For (2). Define $\varphi: N(H) \rightarrow \text{Aut}_\pi(\tilde{X})$ by sending $[\gamma]$ to the deck transformation τ taking $\tilde{x}_0$ to $\tilde{x}_1$, in the notation above. Let's show φ is a homomorphism. If γ' is another loop corresponding to the deck transformation τ' taking $\tilde{x}_0$ to $\tilde{x}'_1$, then $\gamma \cdot \gamma'$ lifts to $\tilde{\gamma} \cdot (\tau(\tilde{\gamma}'))$, a path from $\tilde{x}_0$ to $\tau(\tilde{x}'_1) = \tau\tau'(\tilde{x}_0)$, so $\tau\tau'$ is the deck transformation corresponding to $[\gamma][\gamma']$. By the proof of (1) one has φ is surjective. The kernel of φ consists of classes $[\gamma]$ lifting to loops in $\tilde{x}$, which are exactly the elements of $\pi_*(\pi_1(\tilde{X}, \tilde{x}_0)) = H$. $\square$

Corollary 2.7. Let X be a topological space and $\pi: \tilde{X} \rightarrow X$ be its universal covering space. Then the quotient space $\tilde{X}/\pi_1(X)$ is homeomorphic X .

Proof. It follows from Proposition 2.4.2 since $\pi_1(X) \cong \text{Aut}_\pi(\tilde{X})$ if $\tilde{X}$ is the universal covering. $\square$

2.5. Covering of manifold.

Lemma 2.5.1. Let X be a topological space admitting a countable open covering $\{U_i\}$ such that each set U_i is second countable in the subspace topology. Then X is second countable.

Proof. Let $\mathcal{B}_\alpha$ be a countable base for U_α . Its members are by definition open in U_α , and as all U_α are open in X , these sets are also open in X . So $\mathcal{B} = \bigcup_\alpha \mathcal{B}_\alpha$ is a countable family of open sets in X . Suppose that $x \in X$ and V is open in X with $x \in V$. Then $x \in U_\beta$ for some index β . Now apply the definition of a base to see that for some $B \in \mathcal{B}_\beta$ we have $x \in B \subseteq V \cap U_\beta$. This $B \in \mathcal{B}$ and $x \in B \subseteq V$. This shows that $\mathcal{B}$ is a countable base for X . $\square$

Theorem 2.6. Suppose M is a topological n -manifold and let $\pi: \widetilde{M} \rightarrow M$ be a covering map. Then $\widetilde{M}$ is a topological n -manifold.

Proof. Since π is a local diffeomorphism and M is locally Euclidean, one has $\widetilde{M}$ is also locally Euclidean. Now let's show $\widetilde{M}$ is Hausdorff, let $\tilde{x}_1, \tilde{x}_2$ be two distinct points in $\widetilde{M}$. If $\pi(\tilde{x}_1) = \pi(\tilde{x}_2)$ and $U \subseteq M$ is an evenly covered open subset containing $\pi(\tilde{x}_1)$, then the component of $\pi^{-1}(U)$ containing $\tilde{x}_1$ and $\tilde{x}_2$ are disjoint open subsets of $\widetilde{M}$ that separate $\tilde{x}_1$ and $\tilde{x}_2$. If $\pi(\tilde{x}_1) \neq \pi(\tilde{x}_2)$, there are disjoint open subsets $U_1, U_2 \subseteq M$ containing $\pi(\tilde{x}_1)$ and $\pi(\tilde{x}_2)$ since M is Hausdorff, and then $\pi^{-1}(U_1)$ and $\pi^{-1}(U_2)$ are disjoint open subsets of $\widetilde{M}$ containing $\tilde{x}_1$ and $\tilde{x}_2$, and thus $\widetilde{M}$ is Hausdorff.

To see $\widetilde{M}$ is second countable, firstly note that each fiber of π is countable since by Corollary 2.4 one has the degree of covering less than or equal $|\pi(M, x)|$, and by Theorem 1.2 one has the fundamental group of a topological manifold is countable.

The collection of all evenly covered open subsets is an open covering of M , and therefore has a countable subcover $\{U_i\}$. For any given i , each component of $\pi^{-1}(U_i)$ contains exactly one point in each fiber over U_i , so $\pi^{-1}(U_i)$ has countably many components. The collection of all components of all sets of the form $\pi^{-1}(U_i)$ is a countable open covering of $\widetilde{M}$. Since each such component is second countable, by Lemma 2.5.1 one has $\widetilde{M}$ is also second countable. $\square$

3. GROUP ACTION

3.1. Continuous action.

Definition 3.1.1 (group action). Let G be a group and S be a set. A left G -action on S is a function

$$\theta: G \times S \rightarrow S$$

satisfying the following two axioms:

- (1) $\theta(e, s) = s$, where $e \in G$ is the identity element.
- (2) $\theta(g_1, \theta(g_2, s)) = \theta(g_1 g_2, s)$, where $g_1, g_2 \in G$.

For convenience we denote $\theta(g, s) = gs$ for $g \in G, s \in S$.

Definition 3.1.2 (G -set). Let G be a group. A set S endowed with a left (or right) G -action is called a left (or right) G -set.

Definition 3.1.3 (orbit). An orbit of a group action is the set of all images of a single element under the action by different group elements.

Definition 3.1.4. Let G be a group and S be a left G -set.

- (1) For $g \in G$, if $gs = s$ for some $s \in S$ implies $g = e$, then the group action is called free.
- (2) For $g \in G$, if $gs = s$ for all $s \in S$ implies $g = e$, then the group action is called effective.
- (3) If for arbitrary $s_1, s_2 \in S$, there exists $g \in G$ such that $gs_1 = s_2$, then the group action is called transitive.

Remark 3.1.1. If a group action is free, then it is effective, but converse statement may not hold.

Definition 3.1.5 (isotropy group). Let G be a group and S be a left G -set. For any $s \in S$, the isotropy group of s , denoted by G_s , is the set of all elements of G that fix s , that is

$$G_s = \{g \in G \mid gs = s\}$$

Remark 3.1.2. It is clear to see the action is free if and only if the isotropy group of every point is trivial.

Definition 3.1.6 (act by homeomorphisms). Let G be a group and X be a topological space. The group G is called acting X by homeomorphisms, if G acts on X , and for every $g \in G$, the map $x \mapsto gx$ is a homeomorphism.

Definition 3.1.7 (topological group). A group is called a topological group, if it is a topological space such that the multiplication and the inversion are continuous.

Definition 3.1.8 (continuous action). Let X be a topological space and G a topological group. A continuous G -action on X is given by the following data:

- (1) G acts on X by homeomorphisms.
- (2) The map $G \times X \rightarrow X$ given by $(g, x) \mapsto gx$ is continuous.

Lemma 3.1.1. Let X be a topological space and G a group acting on X by homeomorphisms. Then the quotient map $\pi: X \rightarrow X/G$ is an open map.

Proof. For any $g \in G$ and any subset $U \subseteq X$, the set $gU \subseteq X$ is defined as

$$gU = \{gx \mid x \in U\}$$

If $U \subseteq X$ is open, then $\pi^{-1}(\pi(U))$ is the union of all sets of the form gU as g ranges over G . Since $p \mapsto gp$ is a homeomorphism, each set is open, and therefore $\pi^{-1}(\pi(U))$ is open in X . Since π is a quotient map, this implies $\pi(U)$ is open in X/G , and therefore π is an open map. $\square$

3.2. Proper action.

Definition 3.2.1 (proper). Let X be a topological space and G a topological group. A continuous G -action on X is called proper if the continuous map

$$\begin{aligned} \Theta: G \times X &\rightarrow X \times X \\ (g, x) &\mapsto (gx, x) \end{aligned}$$

is proper, that is, the preimage of a compact set is compact.

Lemma 3.2.1. Let X, Y be topological spaces and $\pi: X \rightarrow Y$ be an open quotient map. Then Y is Hausdorff if and only if the set $\mathcal{R} = \{(x_1, x_2) \mid \pi(x_1) = \pi(x_2)\}$ is closed in $X \times X$.

Proposition 3.2.1. Let X be a topological space and G a topological group acting on X continuously. If the action is also proper, then the orbit space is Hausdorff.

Proof. Let $\Theta: G \times X \rightarrow X \times X$ be the proper map $\Theta(g, x) = (gx, x)$ and $\pi: X \rightarrow X/G$ be the quotient map. Define the orbit relation $\mathcal{O} \subseteq X \times X$ by

$$\mathcal{O} = \Theta(G \times X) = \{(gx, x) \mid x \in X, g \in G\}$$

Since proper continuous map is closed, it follows that $\mathcal{O}$ is closed in $X \times X$, and since π is open by Lemma 3.1.1, one has X/G is Hausdorff by Lemma 3.2.1. $\square$

Proposition 3.2.2. Let M be a topological manifold and G a second countable topological group acting on M continuously. The following statements are equivalent.

- (1) The action is proper.
- (2) If $\{p_i\}$ is a sequence in M and $\{g_i\}$ is a sequence in G such that both $\{p_i\}$ and $\{g_i p_i\}$ converge, then a subsequence of $\{g_i\}$ converges.
- (3) For every compact subset $K \subseteq M$, the set $G_K = \{g \in G \mid gK \cap K \neq \emptyset\}$ is compact.

Proof. Along the proof, let $\Theta: G \times M \rightarrow M \times M$ denote the map $(g, p) \mapsto (gp, p)$. For (1) to (2). Suppose Θ is proper, and $\{p_i\}, \{g_i\}$ are sequences satisfying the hypotheses of (2). Let U and V be precompact² neighborhoods of $p = \lim_i p_i$ and $q = \lim_i g_i p_i$. The assumption implies $\Theta(g_i, p_i)$ all lie in compact set $\overline{U} \times \overline{V}$ when i is sufficiently large, so there exists a subsequence of $\{(g_i, p_i)\}$ converges in $G \times M$ since Θ is proper. In particular, this means that a subsequence of $\{g_i\}$ converges in G .

For (2) to (3). Since G is second countable, it suffices to show it is sequential compact. Let K be a compact subset of M , and suppose $\{g_i\}$ is any sequence

²A set is called precompact, if its closure is compact.

in G_K . This means for each i , there exists $p_i \in g_i K \cap K$, which is to say that $p_i \in K$ and $g_i^{-1} p_i \in K$. By passing to a subsequence twice, we may assume both $\{p_i\}$ and $\{g_i^{-1} p_i\}$ converge, and the assumption implies there exists a convergent subsequence of $\{g_i\}$. Since each sequence of G_K has a convergent subsequence, G_K is compact.

For (3) to (1). Suppose $L \subseteq M \times M$ is compact, and let $K = \pi_1(L) \cup \pi_2(L)$, where $\pi_1, \pi_2: M \times M \rightarrow M$ are the projections onto the first and second factors, respectively. Then

$$\Theta^{-1}(L) \subseteq \Theta^{-1}(K \times K) = \{(g, p) \mid gp \in K, p \in K\} \subseteq G_K \times K$$

By assumption $G_K \times K$ is compact, and thus $\Theta^{-1}(L)$ is compact since it is a closed subset of a compact subset, which implies the action is proper. $\square$

Corollary 3.1. Let M be a topological manifold and G a compact topological group. Then every continuous G -action on M is proper.

Proof. Since G is compact, every sequence in G admits a convergent subsequence, and thus the action is proper by (2) of Proposition 3.2.2. $\square$

3.3. Properly discontinuous action.

Definition 3.3.1 (properly discontinuous). Let G be a group acting on a topological space X by homeomorphisms. The action is called properly discontinuous, if every point $x \in X$ has a neighborhood U such that for each $g \in G$, $gU \cap U = \emptyset$ unless $g = e$.

Lemma 3.3.1. Suppose G be a group acting properly discontinuous on a topological space X . Then every subgroup of G still acts properly discontinuous on X .

Lemma 3.3.2. Let $\pi: \tilde{X} \rightarrow X$ be a covering space. Then $\text{Aut}_\pi(\tilde{X})$ acts on $\tilde{X}$ properly discontinuous.

Proof. For $\tilde{x} \in \tilde{X}$, choose an open neighborhood $\tilde{U}$ of $\tilde{x}$ such that $\pi|_{\tilde{U}}: \tilde{U} \rightarrow U$ is a homeomorphism. If $g(\tilde{U}) \cap \tilde{U} \neq \emptyset$ for some $g \in \text{Aut}_\pi(\tilde{X})$, then $g\tilde{x}_1 = \tilde{x}_2$ for some $\tilde{x}_1, \tilde{x}_2 \in \tilde{U}$. Since $\pi(\tilde{x}_1) = \pi(g\tilde{x}_1) = \pi(\tilde{x}_2)$ and $\pi|_{\tilde{U}}$ is injective, we have $\tilde{x}_1 = \tilde{x}_2$. Thus g has a fixed point, so $g = e$ by Proposition 2.4.1. $\square$

Proposition 3.3.1. Suppose G is a discrete topological group acting continuously and freely on a topological manifold M . The action is proper if and only if the following conditions both hold.

- (1) G acts on M properly discontinuous.
- (2) If $p, p' \in M$ are not in the same orbit, then there exist neighborhoods V of p and V' of p' such that $gV \cap V' = \emptyset$ for all $g \in G$.

Proof. Firstly, suppose that the action is free and proper and let $\pi: M \rightarrow M/G$ denote the quotient map. By Proposition 3.2.1, the orbit space M/G is Hausdorff. If $p, p' \in M$ are not in the same orbit, we can choose disjoint neighborhoods W of $\pi(p)$ and W' of $\pi(p')$, and then $V = \pi^{-1}(W)$ and $V' = \pi^{-1}(W')$ satisfy the conclusion of condition (2). To show G acts on M properly discontinuous, we need to show for each $p \in M$, there exists an open neighborhood U of p such that $gU \cap U = \emptyset$ unless $g = e$. Let V be a precompact neighborhood of p .

By Proposition 3.2.2, the set $G_{\bar{V}}$ is a compact subset of G , and hence finite because G is discrete, so we write $G_{\bar{V}} = \{e, g_1, \dots, g_m\}$. Shrinking V if necessary, we may assume that $g_i^{-1}p \notin \bar{V}$ for $i = 1, \dots, m$. Consider open subset

$$U = V \setminus (g_1\bar{V} \cup \dots \cup g_m\bar{V})$$

It is clear $gU \cap U = \emptyset$ unless $g = e$.

Conversely, assume that (1) and (2) hold. Suppose $\{g_i\}$ is a sequence in G and $\{p_i\}$ is a sequence in M such that $p_i \rightarrow p$ and $g_i p_i \rightarrow p'$. If p and p' are in different orbits, there exist neighborhoods V of p and V' of p' as in (2), but for large enough i , we have $p_i \in V$ and $g_i p_i \in V'$, which contradicts the fact that $g_i V \cap V' = \emptyset$. This shows p and p' are in the same orbit, so there exists $g \in G$ such that $gp = p'$. This implies $g^{-1}g_i p_i \rightarrow p$. Since G acts on M properly discontinuous, there exists an open neighborhood U such that $gU \cap U = \emptyset$ unless $g = e$. For large enough i , one has p_i and $g^{-1}g_i p_i$ are both in U , and by the choice of U one has $g^{-1}g_i = e$. So $g_i = g$ when i is large enough, which certainly converges. By (2) of Proposition 3.2.2, the action is proper. $\square$

Theorem 3.1. Let M be a topological manifold and $\pi: \tilde{M} \rightarrow M$ be a normal covering space. If $\text{Aut}_\pi(\tilde{M})$ is equipped with the discrete topology, then it acts on $\tilde{M}$ continuously, freely and properly.

Proof. By Proposition 2.4.1 one has $\text{Aut}_\pi(\tilde{M})$ acts on $\tilde{M}$ freely and the action is also continuously since $\text{Aut}_\pi(\tilde{M})$ is equipped with discrete topology. To see the action is properly, it suffices to show the action satisfies the two conditions in Proposition 3.3.1.

- (a) By Lemma 3.3.2, one already has $\text{Aut}_\pi(\tilde{M})$ acts on $\tilde{M}$ properly discontinuous.
- (b) Since $\pi: \tilde{M} \rightarrow M$ is a normal covering, one has the orbit space is homeomorphic to M by Proposition 2.4.2 and thus orbit space is Hausdorff. If $\tilde{x}_1, \tilde{x}_2 \in \tilde{M}$ are in different orbits, we can choose disjoint neighborhoods W of $\pi(\tilde{x}_1)$ and W' of $\pi(\tilde{x}_2)$ since orbit space is Hausdorff, and it follows that $V = \pi^{-1}(W)$ and $V' = \pi^{-1}(W')$ satisfy the second condition.

$\square$

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